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ORIGINAL INVESTIGATION

Training Load and Acute Performance Decrements Following Different Training Sessions

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Purpose: To examine the differences in training load (TL) metrics when quantifying training sessions differing in intensity and duration. The relationship between the TL metrics and the acute performance decrement measured immediately after the sessions was also assessed. ***Methods:*** Eleven male recreational cyclists performed 4 training sessions in a random order, immediately followed by a 3-km time trial (TT). Before this period, participants performed the time TT in order to obtain a baseline performance. The difference in the average power output for the TTs following the training sessions was then expressed relative to the best baseline performance. The training sessions were quantified using 7 different TL metrics, 4 using heart rate as input, 2 using power output, and 1 using the rating of perceived exertion. ***Results:*** The load of the sessions was estimated differently depending on the TL metrics used. Also, within the metrics using the same input (heart rate and power), differences were found. TL using the rating of perceived exertion was the only metric showing a response that was consistent with the acute performance decrements found for the different training sessions. The Training Stress Score and the individualized training impulse demonstrated similar patterns but overexpressed the intensity of the training sessions. The total work done resulted in an overrepresentation of the duration of training. ***Conclusion:*** TL metrics provide dissimilar results as to which training sessions have higher loads. The load based on TL using the rating of perceived exertion was the only one in line with the acute performance decrements found in this study.

Keywords: cycling, quantification, interval, training stress, training impulse

The goal of every professional coach and athlete is to optimize performance to the highest level at the required moment of competition. In achieving this, planning and monitoring of the training load (TL) plays a crucial role. Adequate monitoring of TL could help in maximizing performance through performance modeling[1–3] and at the same time help to diminish the injury risk or the risk of nonfunctional overreaching.[4] It is thus essential that the chosen TL method capture the true load of each training session.

In the 70s, Banister et al[5] developed the training impulse (TRIMP) as a first TL quantification method.[1] Over the years, and with evolving technologies, many other TL methods were developed. A distinction between external and internal TL methods is often made, with the external TL representing the absolute amount of work done by an athlete (eg, distance or speed) and the internal TL representing the psychophysiological response of this external load on the individual (eg, heart rate [HR], blood lactate [La^-], or the rating of perceived exertion [RPE]).[6] The internal TL is often considered superior, since this load represents the metabolic load of exercise and will therefore determine the training adaptations and performance outcome.[6] However, to date, there is still no consensus on which TL methods best represent the true load of a training session.[7–10]

A good TL method provides a measure that is related to an outcome of importance, such as fatigue or performance.[11] In recent years, some studies focused on a linear relationship between TL and performance.[11–13] It is clear that this relationship is too much of an oversimplification.[10] More recent studies have focused on the TL methods used within the fitness-fatigue model.[9,14] It was shown that all TL methods are able to relate training to changes in performance with a reasonably small error.[9,14] However, when calculating the effect of a training session on future performance, differences between TL methods were found.[9] This implies that tapering recommendations would be different by using a different TL metric.

In addition to the above research, focusing on the longitudinal relationship between TL and performance, TL metrics have recently been linked to acute performance decrements (APDs), which should serve as a reflection of acute fatigue following distinct exercise bouts.[7,8] It was found that most TL metrics did not reflect APD induced by the sessions, measured by a change in the power produced over a 5-minute all-out time trial (TT).[7] The APD was highest following the most intense sessions, while TL was highest for the longer low-intensity sessions. Thus, most TL methods seem to overemphasize the duration of a training session, while failing to reflect the load of high-intensity efforts. However, in these studies, the sessions used were not really characteristic for training sessions used in practice, which could limit the transfer of those findings to the field. Of course, the use of APD in order to reflect fatigue is limited, since it only measures the decline in performance, which could be the result of fatigue at different levels and is exercise-mode dependent.[15]

Therefore, in this study, we investigated the relationship between different TL metrics and APD induced by 4 different training sessions that are commonly used in training practice. By varying the sessions as to the intensity and duration, we want to demonstrate which TL methods best reflect the stochastic nature of cycling. It was hypothesized that the load of the training sessions would be estimated differently depending on the TL metric used and that the internal TL metrics would better reflect the APD induced by the training sessions.

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Methods

Subjects

Eleven healthy physically active men (26.9 [4.2] y, 182.8 [4.6] cm, 72.6 [6.9] kg) participated voluntarily for this study. They were all recreationally active as cyclists (1500–5000 km·y^{-1}). The participants were informed about the risks of the study, and written informed consent was obtained before participation. They underwent a medical examination and were declared to be in good health. This study was approved by the Ethical Committee of the Ghent University Hospital (Ghent, Belgium).

Experimental Procedures

Overview. A randomized, crossover, within-subjects design was used whereby participants visited the lab on 7 occasions, separated by at least 48 hours in a 3-week period. Session 1 consisted of a step incremental exercise test, and sessions 2 and 3 were 2 baseline 3-km TTs. During sessions 4 to 7, the participants completed a training session immediately followed by a TT (Figure 1). All tests and training sessions took place in the Sport Science Laboratory—Jacques Rogge (Ghent University, Belgium), under controlled environmental conditions (18–19 °C, 50% relative humidity). The subjects were asked to keep their nutritional intake at the same level for all sessions and were asked to limit caffeine intake preceding the sessions. During all sessions, HR (H10 Sensor, Polar) and power output (Cyclus2 ergometer, RBM Electronics) were continuously monitored in order to quantify TL. Pulmonary gas exchange (oxygen uptake [$\dot{V}O_2$], $\dot{V}CO_2$) was measured breath by breath (Jaeger Oxycon Pro, Viasys Healthcare GmbH), and La^- was sampled (Biosen C-line, EKF Diagnostics GmbH) at regular intervals during every training session.

Exercise Test. The subjects performed a step incremental test to assess their fitness level and to individualize the training sessions. The test started at 80 W, and every 3 minutes, the power output was increased by 40 W. The participants were instructed to keep their preferred cadence constant. The protocol was terminated at voluntary exhaustion, that is, the inability to maintain their preferred cadence minus 10 rpm for more than 5 consecutive seconds, despite strong verbal encouragement.

Peak oxygen uptake was defined as the highest 30-second average achieved during the test. The aerobic threshold (AT) was determined as the first visual rise on the La^-/power output curve above resting values.[16] The anaerobic threshold (ANT) was determined as

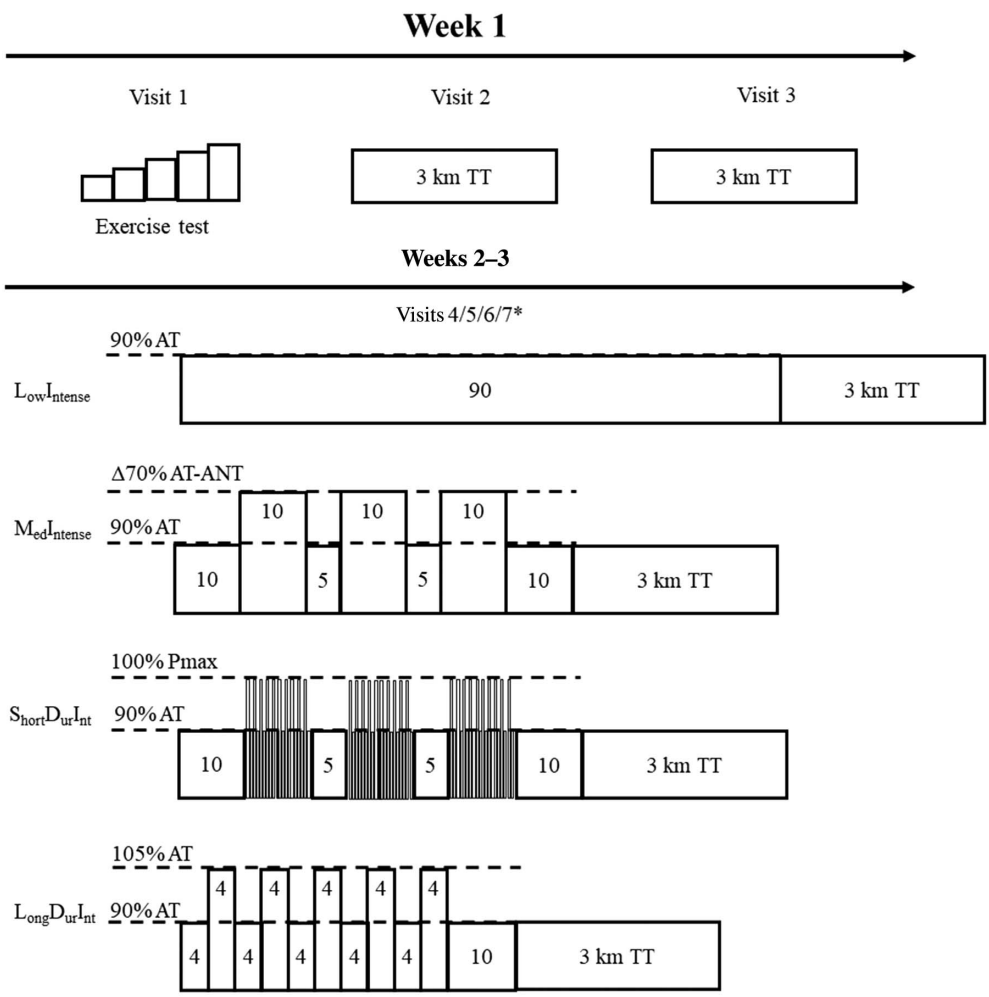

Figure 1 — Overview of the study design. ANT indicates power output at the anaerobic threshold; AT, power output at the aerobic threshold; $L_{ong}D_{ur}I_{nt}$, long-duration interval; $L_{ow}I_{ntense}$, low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; P_{max}, maximal power output produced during exercise test; $S_{hort}D_{ur}I_{nt}$, short-duration interval; TT, time trial. *The training sessions were randomized for visits 4, 5, 6, and 7.

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the second deflection point in the La^-/power output curve. Both thresholds were determined by 3 independent researchers. Next to La^-, different variables (HR, $\dot{V}O_2$, and RPE) were also taken into account when determining this threshold.[17] Consensus was sought when discrepancies occurred. The $\dot{V}O_2$ corresponding with the thresholds was also determined. Maximal HR (HR_{max}) was determined as the highest HR obtained during this test. Resting HR was defined as the lowest 5-second average HR when subjects were lying in a supine position for 10 minutes in a quiet room.

Time Trials. The gear settings for the 3-km TT were selected for every participant so that the power output during the first 250 m of every TT equaled 105% of the maximal power output (P_{max}) achieved during the exercise test at their preferred cadence. The subjects had to keep the cadence constant for the first 250 m and could thereafter cycle freely; this was done in order to help the participants pace in a similar manner over the different TTs. The first 2 TTs were preceded by a short warm-up of 6 minutes wherein they had to perform 3 blocks of 1 minute at the power output corresponding to ANT. The other TTs were performed immediately after the training session. The subjects were encouraged to complete the TT as fast as possible. The only feedback they received was cadence and remaining distance.

Training Sessions. Four different training sessions were completed. All were equalized for Lucia's TRIMP based on power output (luTRIMPpo) with a total luTRIMPpo of 90 (see the "Data Analysis" section). The training sessions were completed in a randomized order (Figure 1). For every session, the warm-up and all recovery blocks were equal to 90% of the power output associated with AT.

- The short-duration interval ($S_{hort}D_{ur}I_{nt}$) consisted of 3 blocks of 10 minutes. During these blocks, a constant repetition of 30 seconds at P_{max} and a 30-second recovery was performed. Recovery between the blocks was again 5 minutes, and the warm-up and cooldown were 10 minutes. The total duration was 60 minutes.
- The long-duration interval ($L_{ong}D_{ur}I_{nt}$) consisted of 5 blocks of 4 minutes at 105% ANT, with a 4-minute recovery. The warm-up was 4 minutes, and the cooldown was 10 minutes. The total duration was 50 minutes.
- The medium-intensity training ($M_{ed}I_{ntense}$) consisted of 3 blocks of 10-minute power output in between the thresholds (at $\Delta 70\ [ANT - AT] \times 0.7 + AT$) interspersed with 5 minutes of recovery. The warm-up and cooldown were each 10 minutes. The total duration was 60 minutes.
- The low-intensity training ($L_{ow}I_{ntense}$) consisted of 90 minutes of the power output associated with AT.

All training sessions were immediately followed by a TT in order to determine APD. The TT was performed immediately after a 10-minute cooldown in order to ensure the measurement of APD caused by the whole training rather than measuring the effect of the last intensity block. After every training segment, the participants were asked to give an RPE score using the Category-Ratio (CR-10) scale,[18] and a blood sample was taken from the fingertip and analyzed for La^- (Biosen C-line, EKF Diagnostics GmbH). Prior to every TT, the participants were asked to score their motivation level on a 0 to 10 scale. The APD was defined as the percentage change in the average power output of the TTs relative to their best baseline TT (baseline).

Data Analysis

Eight different TL methods were used, 5 internal and 3 external TL metrics.

All training sessions were equal for luTRIMPpo where the time under AT was multiplied by 1, the time between the thresholds by 2, and the time above ANT by 3. This is a slightly adapted version from the original luTRIMP that has been used before.[10,11] Training Stress Score (TSS) was selected as a second external TL measure. TSS is calculated using the following formula[19]:

$$\mathbf{TSS} = (\mathbf{t} \times \mathbf{NP} \times \mathbf{IF}) / (\mathbf{FTP} \times 3600) \times 100,$$

where t is the time, NP is normalized power, and IF is the intensity factor (= NP/FTP). FTP is an individual's functional threshold power. For this study, FTP was assumed to be equal to the determined ANT.

As a last external TL measure, the total work done (TWD), or mechanical energy spent (in kilojoules), was calculated from the power output.

Four HR-based TL methods were used. The Banister TRIMP (bTRIMP) was calculated using the training duration, average HR, and an intensity factor (IF) using the following equation[5]:

$$\mathbf{bTRIMP} = \mathbf{Duration\ training\ (in\ minutes)} \times \mathbf{\Delta HR} \times 0.64\mathbf{e}^{1.92\mathbf{x}},$$

where $\mathbf{\Delta}HR = (HR_{mean} - \text{resting HR})/(HR_{max} - \text{resting HR})$, e is the exponential constant, 1.92 and 0.64 are generic constant for males, and $x = \mathbf{\Delta}HR$.

Five predefined zones (zone 1, 50%–59% HR_{max}, IF = 1; zone 2, 60%–69% HR_{max}, IF = 2; zone 3, 70%–79% HR_{max}, IF = 3; zone 4, 80%–89% HR_{max}, IF = 4; zone 5, 90%–100% HR_{max}, IF = 5) were used to calculate the Edwards TRIMP. The time spent in each zone was multiplied by the respective IF and then summated in order to compute the total Edwards TRIMP score.[20]

For the calculation of the Lucia TRIMP based on HR, 3 predefined HR zones were used. Zone 1 (IF = 1) was defined as below AT, zone 2 (IF = 2) between AT and ANT, and zone 3 (IF = 3) above ANT. Again, the time in each zone was multiplied by its respective IF and then summed to provide a total Lucia TRIMP based on HR.[21]

Calculation of the individualized TRIMP (iTRIMP) was performed using the individual's La^-–$\mathbf{\Delta}$HR relationship as an exponential IF using the best fitting method. This exponential IF was used to weight every measured HR.[12]

The TL was also calculated according to the method of Foster,[18] as a subjective measure of internal TL (TLrpe). After every session, the subjects were asked to rate their session RPE (sRPE) using the CR-10 scale. TLrpe was then calculated by multiplying this sRPE by the session duration.

Statistical Analysis

The data are presented as mean (SD). A repeated-measures analysis of variance was used to analyze differences within TL methods for the different training sessions and to check for differences between the different TTs. Effect sizes (ESs; Cohen d) were calculated for differences between the TTs, and values were interpreted as follows: 0 to 0.19, trivial; 0.20 to 0.59, small; 0.6 to 1.19, moderate; 1.20 to 1.99, large; and ≥2.00 very large. All statistical analyses were performed with SPSS Statistics (version 28), and statistical significance was set at $P < .05$.

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Results

Subject Overview

The P_{max} from the step incremental exercise test was 312 (34) W with a corresponding peak oxygen uptake of 4.1 (0.5) L·min^{-1} (57.3 [7.1] mL·kg^{-1}·min^{-1}). The power output associated with the AT and ANT were 149 W (22) and 231 W (23), respectively.

Training Sessions

The luTRIMPpo showed no differences between training sessions, since the sessions were designed to be equal for luTRIMPpo.

Within the TL metrics, there were significant ($P < .05$) differences between the different training sessions (Figure 2). The main findings are that $S_{hort}D_{ur}I_{nt}$ was highest for iTRIMP, TSS, and TLrpe ($P < .05$). $S_{hort}D_{ur}I_{nt}$ was equal to $M_{ed}I_{ntense}$ and $L_{ow}I_{ntense}$ for bTRIMP ($P = .112$; $P = .055$), Edwards TRIMP ($P = .172$; $P = .838$), and Lucia TRIMP based on HR ($P = .106$; $P = .163$), but higher than $L_{ong}D_{ur}I_{nt}$ for all of these methods ($P < .05$). $L_{ow}I_{ntense}$ was lowest only for TSS and iTRIMP ($P < .05$). TWD was the only metric where all sessions differed from each other ($P < .01$), and this in the following order: $L_{ow}I_{ntense} > S_{hort}D_{ur}I_{nt} > M_{ed}I_{ntense} > L_{ong}D_{ur}I_{nt}$.

An overview of the psychophysiological data registered during the different training sessions is given per training (Figure 3). $S_{hort}D_{ur}I_{nt}$ and $L_{ong}D_{ur}I_{nt}$ induced a higher maximal ($P < .001$) and end ($P < .05$) La^{-} value compared with $M_{ed}I_{nt}$ and $L_{ow}I_{ntense}$, with $M_{ed}I_{nt}$ also inducing higher values compared with $L_{ow}I_{ntense}$ ($P < .001$). Furthermore, the relative contribution of time spent at a $\dot{V}O_2$ above ANT was higher in $S_{hort}D_{ur}I_{nt}$ and $L_{ong}D_{ur}I_{nt}$ compared with $M_{ed}I_{ntense}$ ($P = .011$ and $P = .001$) and $L_{ow}I_{ntense}$ ($P = .009$ and $P = .001$) (Table 1).

Time Trials

Only the TT performed after the $S_{hort}D_{ur}I_{nt}$ training showed a significant decrease in performance (5.3% [3.9%], $P = .003$; ES: 1.26, large) compared with the baseline. The TTs after $L_{ong}D_{ur}I_{nt}$ (+12.8 [4.1] W, $P = .012$; ES: 0.88, moderate), $M_{ed}I_{ntense}$ (+ 12.8 [5.3] W, $P = .036$; ES = 0.88, moderate), and $L_{ow}I_{ntense}$ (+ 15.6 [5.8] W, $P = .022$; ES: 1.04, moderate) were all better than for $S_{hort}D_{ur}I_{nt}$, but did not differ from each other and from the baseline. An overview of the individual differences is given in Figure 4. An overview of the group-averaged power output for every 500-m segment is shown in Figure 5, where differences in pacing can be seen. Before the start of every TT, motivation was equal; HR_{max} and RPE from the TTs did not differ either (Table 2).

Discussion

The aim of this study was to assess the relationship between existing TL methods and the APD induced by 4 commonly used training sessions in cycling. If this relationship is clear, it could help coaches and sport scientists in answering the questions as to which TL methods should be used in order to monitor athletes in the best way, as they best represent the load of sessions. Already, here it should be mentioned that the use of APD is only one of the methods that can partly explain the usefulness of different TL methods. This method is not superior to other methods trying to elucidate the TL concept, but serves as an addition to other methods, and the findings should be put in this perspective.

First, it is clear that using different TL quantification methods leads to different patterns in the load of the training sessions in this study (Figure 2), as well as when comparing within the internal and external TL methods. The APD for $S_{hort}D_{ur}I_{nt}$ was larger than for the other training sessions, which is reflected by the TLrpe, iTRIMP, and TSS methods, but not in the rest. These were the only methods that assessed the TL of the $S_{hort}D_{ur}I_{nt}$ session as being higher than for the other training sessions, thus reflecting the APD measured. However, TL calculated by iTRIMP and TSS also show a smaller load for the $L_{ow}I_{ntense}$ session compared with $M_{ed}I_{ntense}$ and $L_{ong}D_{ur}I_{nt}$, which is not compatible with the APD. This implies that for those 2 methods, the intensity of the training sessions is weighted even more than necessary. Also, in an earlier

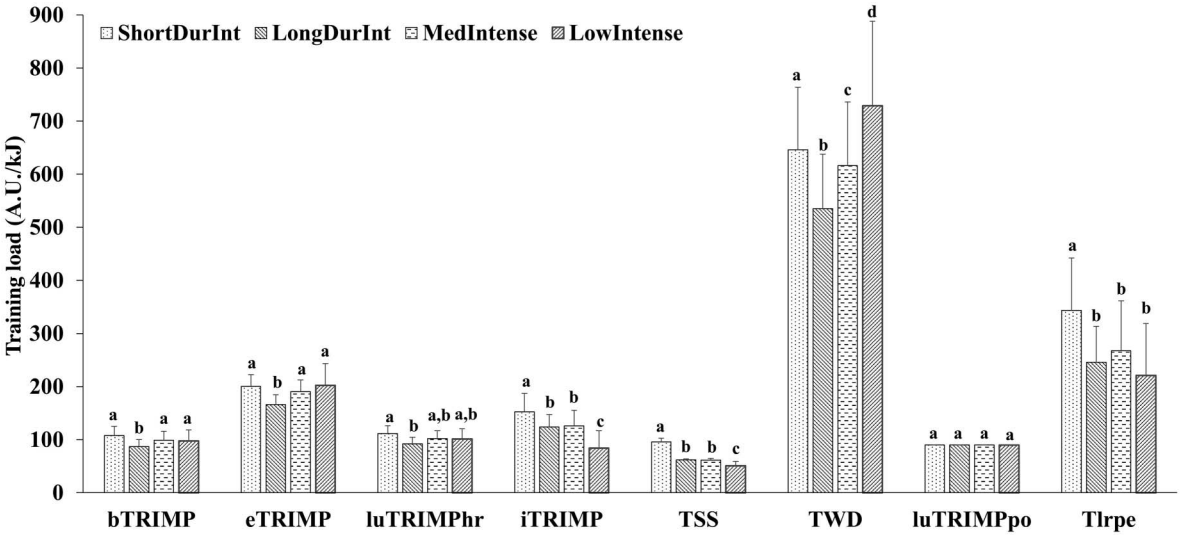

Figure 2 — Overview of the differences between TL values. Identical letters indicate that there are no significant differences between training sessions within the TL method. Different classifications of training sessions can be seen depending on the TL method used. A.U. indicates arbitrary units; $L_{ong}D_{ur}I_{nt}$, long-duration interval training; $L_{ow}I_{ntense}$, low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; $S_{hort}D_{ur}I_{nt}$, short-duration interval training; TL, training load; TL_{rpe}, TL using rating of perceived exertion; TRIMP, training impulse; TSS, Training Stress Score; TWD, total work done.

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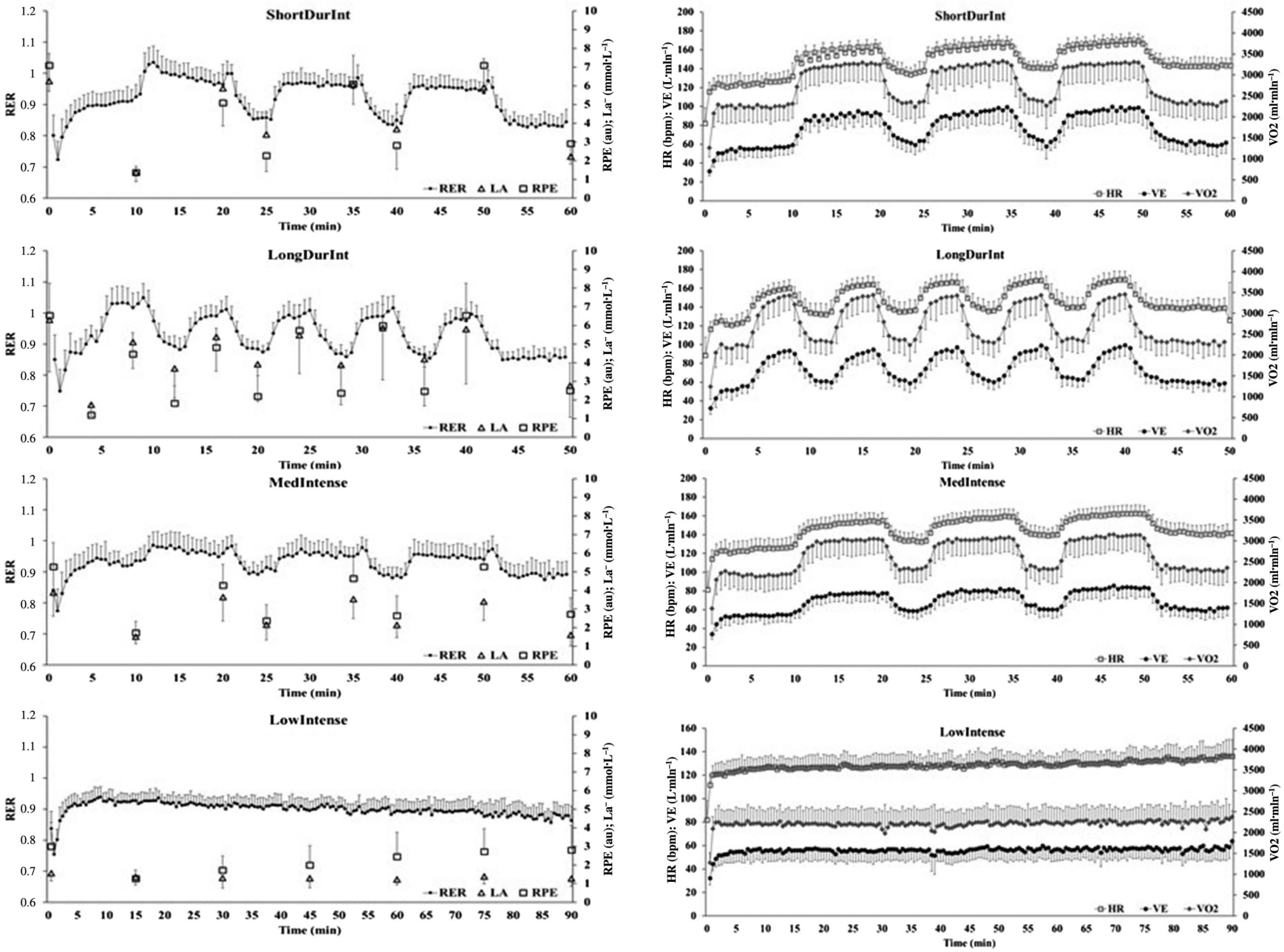

Figure 3 — Overview of the training data registered. Au indicates arbitrary units; bpm, beats per minute; HR, heart rate; La^-, blood lactate; $L_{ong}D_{ur}I_{nt}$, long-duration interval; $L_{ow}I_{ntense}$, low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; RER, respiratory exchange ratio; RPE, rating of perceived exertion; $S_{hort}D_{ur}I_{nt}$, short-duration interval; $\dot{V}E$, ventilation; $\dot{V}O_2$, oxygen uptake.

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Table 1 Overview of Differences in Physiological Variables Measured

	$S_{hort}D_{ur}I_{nt}$	$L_{ong}D_{ur}I_{nt}$	$M_{ed}I_{ntense}$	$L_{ow}I_{ntense}$
La^- max, mmol·L^{-1}	6.25^a (1.47)	6.3^a (2.8)	3.89^b (1.27)	1.61^c (0.4)
La^- end, mmol·L^{-1}	2.23^a (0.83)	2.77^a (1.71)	1.62^b (0.61)	1.28^c (0.42)
$\dot{V}O_2$-zone 1, %	35.1^a (9.7)	37.5^a (14.6)	36.7^a (10.3)	93.3^b (7.5)
$\dot{V}O_2$-zone 2, %	46.7^a (14.9)	42.7^a (8.8)	61.3^b (10.3)	6.7^c (7.5)
$\dot{V}O_2$-zone 3, %	18.2^a (17.7)	19.8^a (13.0)	2.0^b (1.8)	0.0^c (0.0)
$\dot{V}O_2$ total, L	165.3^a (22.6)	135.7^b (17.0)	158.3^c (19.7)	197.9^d (30.9)

Abbreviations: La^- end, La^- measured at the end of the training session; La^- max, highest La^- measured during the training; $L_{ong}D_{ur}I_{nt}$, long-duration interval; $L_{ow}I_{ntense}$, low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; $S_{hort}D_{ur}I_{nt}$, short-duration interval; $\dot{V}O_2$-zones 1/2/3, the percentage of time spent under the $\dot{V}O_2$ corresponding with AT, the percentage of time spent between the thresholds and the percentage of time spent above the $\dot{V}O_2$ corresponding with ANT; $\dot{V}O_2$ total, total volume of consumed oxygen. Mean (SD) are presented for the following: La^- max, La^- end, $\dot{V}O_2$-zones 1/2/3, and $\dot{V}O_2$ total. Note: Identical letters in superscript indicate there is no statistical difference ($p > .05$) for this metric between training session.

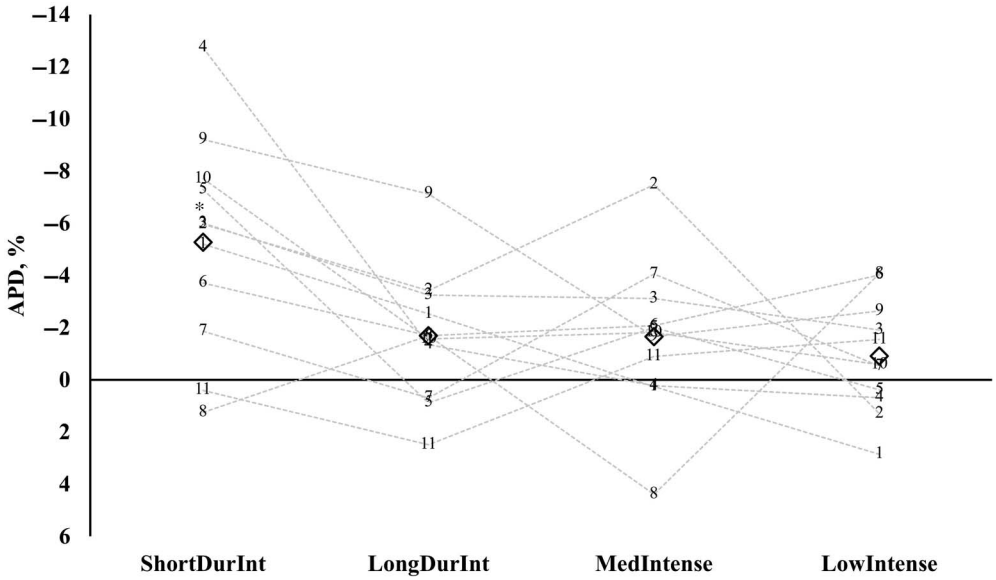

Figure 4 — Overview of the individual differences for the APD following the different training sessions compared with baseline. Individuals are represented by individual numbers. The diamond shapes represent the group average. APD indicates acute performance decrements; $L_{ong}D_{ur}I_{nt}$, long-duration interval training; $L_{ow}I_{ntense}$, low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; $S_{hort}D_{ur}I_{nt}$, short-duration interval training, *Statistical difference (p<0.05) from other training sessions.

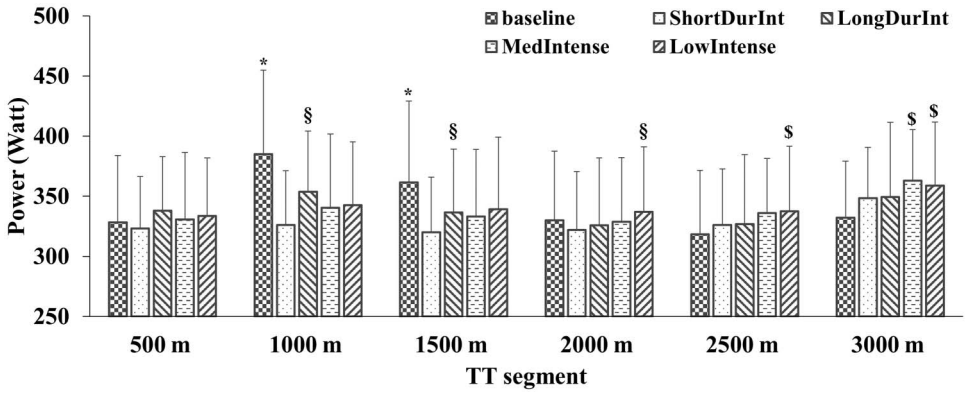

Figure 5 — Overview of the group-averaged power per 500-m segment during the different time trials following the different training sessions. It is clear that the pacing differs from baseline in the first half of the TT. Starting from the 2000-m segment, the $L_{ow}I_{ntense}$ TT starts to differ from the $S_{hort}D_{ur}I_{nt}$ TT. At the last 500 m more power is produced for the $L_{ow}I_{ntense}$ and $M_{ed}I_{ntense}$ TT than for baseline. $L_{ow}I_{ntense}$ indicates low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; $S_{hort}D_{ur}I_{nt}$, short-duration interval training; TT, time trial. *Significant difference from all other TTs at that segment. §Significant difference from the $S_{hort}D_{ur}I_{nt}$ TT at that segment. $Significant difference from bestTT at that segement.

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Table 2 Overview of Time Trial Characteristics

	Best	$S_{hort}D_{ur}I_{nt}$	$L_{ong}D_{ur}I_{nt}$	$M_{ed}I_{ntense}$	$L_{ow}I_{tense}$
Mean power, W	342.6 (15.9)	323.8* (13.9)	336.6 (15.3)	336.6 (15.3)	339.4 (15.8)
HR_{max}, bpm	187 (9.8)	187 (9.6)	188 (9.5)	188 (8.9)	188 (8.6)
RPE, 0–10	9.72 (0.62)	9.8 (0.39)	9.8 (0.39)	10 (0.0)	9.9 (0.16)
Motivation, 0–10	7.5 (0.63)	6.9 (0.5)	7.1 (0.4)	7.3 (0.5)	7.3 (0.4)

Abbreviations: bpm, beats per minute; HR, heart rate; HR_{max}, maximal HR; $L_{ong}D_{ur}I_{nt}$, long-duration interval training; $L_{ow}I_{ntense}$, low-intensity training; $M_{ed}I_{ntense}$, medium-intensity training; RPE, rating of perceived exertion; $S_{hort}D_{ur}I_{nt}$, short-duration interval training. Note: Mean power during the time trials, maximal measured HR during the time trials, RPE given at the end of the time trials and motivation scored before the start of the time trials, all values are presented as mean (SD).
*Significant difference from other TTs ($P < .05$).

study, it was already shown that TSS is influenced more by intensity than other TL metrics (TWD, luTRIMP, and sRPE) in elite cyclists.[22]

For all other internal TL methods, which were all based on HR, the $L_{ong}D_{ur}I_{nt}$ training showed the smallest load, and at the same time, $L_{ow}I_{ntense}$ had a load as high as the more intense sessions. Since $L_{ong}D_{ur}I_{nt}$ is the shortest session and $L_{ow}I_{ntense}$ is the longest session, this indicates that, for those TL metrics, the duration is clearly emphasized more than intensity, which is incoherent with the APD found in this study. It is thus paramount that coaches understand that, even based on the same input, some TL methods put more emphasis on intensity, while others emphasize duration to a greater extent. The fact that intensity of training plays a crucial role in the TL metrics was also shown before in elite cyclists.[22] Here it was shown that the relationship between TWD and TSS changes when the intensity of training is higher, with TSS emphasizing intensity more.

The most remarkable difference between the training sessions in the present study was found for TWD where $L_{ow}I_{intense}$ was scored highest, followed by $S_{hort}D_{ur}I_{nt}$, then $M_{ed}I_{ntense}$, and then the $L_{ong}D_{ur}I_{nt}$ training. Our results thus highlight that the way kilojoules are spent/produced is more important than the total amount of TWD when trying to assess the load of a training session. This is supported by the fact that the modalities of training (intensity) seem to have a larger effect on APD than the TWD, both in this study and in previous research using a similar approach.[7] This is especially noteworthy to mention, since the TWD is often used in studies and in the field to compare races,[23] TL methods,[22,24] or training programs.[25,26] Still, the same pattern was found for TWD and the total volume of $\dot{V}O_2$ during training sessions, suggesting that TWD and $\dot{V}O_2$ are related to each other. Nevertheless, according to our results, it would be better to match efforts based on TLrpe, at least in study designs where similar modalities are used, as in the present study.

Hypothetically, for this group of subjects, this would imply that, on average, a $L_{ow}I_{ntense}$ session of 2 hours and 17 minutes (2.5 [average sRPE for $L_{ow}I_{intense}$] × 137 min = 343 arbitrary unit) would result in a load similar to the average load of our subjects for the 60-minute $S_{hort}D_{ur}I_{nt}$ (344 arbitrary unit) training used in this study, and these training sessions would then be matched for TL in a better way than when using TWD. However, here, the rather strong assumption is made that the sRPE (2.5) of the $L_{ow}I_{ntense}$ session would not be different after a session of 137 minutes compared with 90 minutes at the same intensity and would thus merely act as an IF. Still, it was shown before that sRPE does not only merely act as an IF, but also includes a duration effect.[27] This effect is also visible for this study in Figure 3 where RPE (assessed every 15 min during training) rises throughout the $L_{ow}I_{ntense}$ session without a rise in power output and $\dot{V}O_2$ (~intensity). So the assumption made before that sRPE would be similar after a longer session of 137 minutes at the same intensity should be made with some caution, since it is likely that the rise in RPE at the same intensity during a session could be reflected in the sRPE assessed after a session.

The fact that there was only a significant APD found for the $S_{hort}D_{ur}I_{nt}$ training was somewhat surprising, especially when compared with the results of a recent study where APDs of 4% to 18% were found after shorter exercise bouts, both maximal and submaximal in nature.[7] But also there, submaximal exercise bouts led to significantly smaller APDs, suggesting that the effort is of greater importance than the duration. Since none of the present training sessions were maximal in nature, and at the end of every training session, there was a 10-minute cool-down period, the results may not be that surprising after all. Also, when taking a closer look at the pacing used during the TTs, a clear difference can be seen for the baseline TT compared with the others, despite controlling the pacing in the first 250 m of the TT. One explanation for this could be psychological in nature, with participants being more cautious when starting the TT after a training session. It is, however, also possible that some peripheral fatigue was present, inducing more afferent feedback, which could have influenced the power output in the first part of the TTs. Earlier studies already demonstrated that peripheral fatigue leads to different pacing strategies, without necessarily reducing the average power output.[28]

Furthermore, the selection of the performance measure is also crucial in determining the relationship between TL and APD. The TT used in the present study lasted approximately 4 to 5 minutes, depending on the participant. Since a maximal effort of that duration is typically ~80% to 85% aerobic in nature,[29] the relationship shown in this study can only be translated to APD in this type of exercise. Furthermore, during this high-intensity trial (109% [5%] P_{max}), a large proportion of type II fibers will be activated. In contrast, during the $L_{ow}I_{ntense}$ session, mostly type I fibers will have been activated, partly depleting the glycogen in those muscle fibers without hampering the type II fibers[15] that are activated for the TT, thus preserving much of the performance capacity, which may hide a fatiguing effect at other levels. The data from our lab (unpublished data) also show that using a 15-second isokinetic sprint does not show any APD following these training sessions, indicating again that the choice of the performance assessment plays an important role in further elucidating this relationship.

Together, the findings of this study show that the duration of training sessions is overemphasized in most, but not all, TL metrics (ie, the fact that LIT is equal or higher in load than the high-intensity sessions), with TLrpe showing a similar pattern to the APD found. Although the training sessions used in this study are

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also often used in the field, they are still somewhat limited, since these are sessions performed indoors. Future studies should thus investigate this relationship again in longer training sessions, with varying intensity and with a more stochastic nature, as to represent the typical outdoor training sessions and races. Also, different performance measures should be used, and in this regard, studies assessing the relationship between TL metrics and measures of central and peripheral fatigue would be of great benefit in further elucidating the meaning of TL methods.

Practical Applications

It is important for practitioners to understand that using a different TL quantification method leads to differences in the load calculated, even when using the same input variable (HR for TRIMPs, power output for TSS and TWD). It is also important to understand that these differences in calculation may often be hidden in practice when looking at correlations between TL methods for a lot of data and much longer training sessions[10,22,24]; still, this does not mean that these differences are to be overlooked. Also, we discourage the use of TWD as a basis for matching and comparing efforts between or within groups. We also want to point out that the use of APDs to assess the validity of TL metrics should not be seen as being the gold standard method; it is merely one of the possible methods that can provide us with more information on the different TL methods. Therefore, the results of these studies should be complemented with other research that investigates the usefulness of TL methods in different ways.

Conclusion

We conclude that using different TL metrics will lead to a different ranking of training sessions as assessed by the TL. Moreover, some of the TL metrics (TLrpe, iTRIMP, and TSS) were able to reflect the APD induced by the heavier session, while the other metrics overemphasized duration.

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